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Q3 2-Methoxy-1,4-naphthoquinone (MNQ) induces apoptosis of A549 lung adenocarcinoma cells via oxidation-triggered JNK and p38 MAPK signaling pathways

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ABSTRACT

Aim: The compound 2-methoxy-1,4-naphthoquinone (MNQ) was previously shown to be cytotoxic against several cancer cell lines, but its mode of action is poorly understood. In this study, we aimed to explore the molecular mechanism of MNQ-induced cytotoxicity of A549 lung adenocarcinoma cells.

Main methods: The growth inhibition potential of MNQ was analyzed using sulforhodamine B assay, flow cytometry cell cycle analysis and Annexin V apoptosis assay. Oxidative stress was determined using 2',7'-dichlorofluorescein diacetate to measure intracellular reactive oxygen species level and comet assay to measure DNA damage. Western blotting was performed to study the activation of mitogen-activated protein kinase signaling pathways.

Key findings: MNQ induced apoptosis of A549 cells independent of cell cycle arrest, and is mediated by the JNK and p38 MAPK signaling pathways. Further analysis demonstrated that these signaling pathways were stimulated by oxidative DNA damage caused by increased ROS generation in MNQ-treated A549 cells.

Significance: This study is the first to provide an insight into the molecular mechanism of MNQ-induced cytotoxicity of a lung cancer cell, which demonstrates the potential of MNQ as a potential chemotherapeutic drug for lung cancer treatment.

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1. Introduction

Lung cancer causes the highest cancer-related mortality globally [31], with an estimate of 1.59 million deaths reported in the year 2012 [15]. This disease can be generally categorized as either non-small cell lung carcinoma (NSCLC), which comprises 85% of the total lung cancer cases reported, or small cell lung carcinoma (SCLC) which accounts for the remaining 15% [30]. Despite the various treatments available for lung cancer management, the survival rates reported are usually low [1,21,23]. In addition, the current chemotherapeutic regimens are compromised with detrimental side effects [28]. Therefore, this warrants the need for more efficient therapeutic approaches.

Quinones are a class of organic compounds in which many of its members were observed to possess anticancer properties by modulating intracellular processes to trigger the cell death mechanisms. These processes include the MAPK signaling pathways, which are deregulated in different types of cancers to promote cell survival and proliferation while evading growth suppression and cell death [6,13,33,35,39]. However, certain anticancer quinones have been observed to overcome this

aberration by stimulating the MAPKs to activate the apoptotic pathways [7,10,16,29].

In addition, several quinones were also observed to promote cancer cell death in a reactive oxygen species (ROS)-dependent mechanism [3, 8,9,25]. The redox systems in most cancer cells were observed to be aberrant, with higher levels of ROS higher compared to normal cells [42]. This higher but non-lethal level of ROS is believed to play an important role in carcinogenesis as they serve as second messengers to alter the signaling pathways, thus promoting proliferation and survival of cancer cells [5,19]. Nevertheless, it is also known that high concentrations of ROS will completely overwhelm the antioxidant defense mechanism, causing extreme oxidative damage to cellular components and results in cell death [11,24]. In fact, several chemotherapeutic drugs function by inducing ROS generation in cancer cells in order to eliminate them [26,40].

The compound 2-methoxy-1,4-naphthoquinone (MNQ) (Fig. 1) is a quinone extracted from garden balsam (*Impatiens balsamina*). This compound has been observed to exert an anticancer effect against several cancer cell lines, including HepG2 hepatocarcinoma cells, MDA-MB-231 breast cancer cells, and MKN45 gastric adenocarcinoma cells [14,22, 41]. Nevertheless, its mechanism of action is not fully understood. In this study, we investigated the growth inhibitory potential of MNQ on A549

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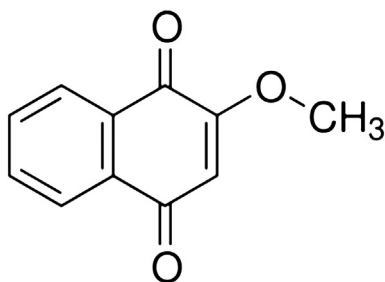


Fig. 1. Chemical structure of 2-methoxy-1,4-naphthoquinone (MNQ).

lung adenocarcinoma cells and elucidated the molecular mechanism of MNQ in inducing A549 lung cancer cell death in regard to MAPK signaling and redox balance.

2. Materials and methods

2.1. Chemicals and reagents

2-Methoxy-1,4-naphthoquinone (MNQ), RPMI 1640, RNase, propidium iodide (PI), N-acetylcysteine (NAC) and dichlorofluorescein diacetate (DCFDA) were purchased from Sigma-Aldrich (St. Louis, MO, USA). Dimethyl sulfoxide (DMSO), JNK inhibitor SP600125 and p38 inhibitor SB203580 were purchased from Calbiochem (San Diego, CA, USA). Primary antibodies against ERK, JNK, p38, and β -actin, as well as HRP-conjugated secondary antibodies were purchased from Santa Cruz Biotechnology (Santa Cruz, CA, USA), while primary antibodies against phospho-ERK, phospho-JNK and phospho-p38 were purchased from Cell Signaling Technology (Beverly, MA, USA). Fetal bovine serum (FBS) was purchased from JR Scientific (Woodland, CA, USA). Annexin V-FITC was purchased from BD Biosciences (San Jose, CA, USA).

2.2. Cell lines and cell culture

A549 lung adenocarcinoma epithelial cell line was purchased from American Type Cell Culture Collection (Manassas, USA) and maintained in RPMI 1640 supplemented with 10% fetal bovine serum. The cells were grown as monolayers at 37 °C and 5% CO₂.

2.3. Sulforhodamine B (SRB) growth inhibition assay

The growth inhibitory potential of MNQ was measured using In Vitro Toxicology Assay Kit Sulforhodamine B-Based (Sigma-Aldrich, St. Louis, MO). Briefly, 5×10^3 A549 cells were seeded in a 96-well cell culture dish and treated with different concentrations of MNQ (0, 0.3125, 0.625, 1.25, 2.5, 5.0, 10, and 20 μ M) for 48 h, or 10 μ M MNQ for 24, 48 and 72 h. The cells were then fixed with TCA and washed and stained with sulforhodamine B (SRB) solution. After washing off unincorporated SRB dye with 1% acetic acid, the protein-bound dye was solubilized with 10 mM Tris and the absorbance was read at 490 nm using the iMark Microplate Reader (Bio-Rad, Hercules, CA).

2.4. Cell cycle analysis

A549 cells were seeded in a 6-well dish and treated with 10 μ M MNQ for 6, 12, 24, and 48 h. The cells were subsequently harvested and fixed with ethanol. The fixed cells were then washed with PBS and resuspended in 20 μ g/ml RNase (Sigma-Aldrich, St. Louis, MO) with 5 μ g/ml PI and incubated in the dark for 30 min. The samples were then analyzed using an Accuri C6 (BD Biosciences, San Jose, CA) flow cytometer. The data obtained were analyzed using the ModFit LT version 4 software (Verity Software House).

2.5. Annexin V apoptosis assay

A549 cells were seeded into a 6-well dish and treated with 10 μ M MNQ for 3, 6, 9, and 12 h. The cells were harvested, washed with PBS and resuspended in $1 \times$ binding buffer (10 mM HEPES, 140 mM NaCl, 2.5 mM CaCl₂) containing Annexin V-FITC and propidium iodide. The cells were then incubated in the dark with Annexin V-FITC and propidium iodide (PI) before being analyzed on the Accuri C6 flow cytometer.

2.6. SDS-PAGE and Western blot

A549 proteins were extracted using M-PER Mammalian Protein Extraction Reagent (Thermo Scientific, Rockford, IL) supplemented with Halt Protease and Phosphatase Inhibitor Cocktail (Thermo Scientific, Rockford, IL) at 4 °C with shaking for 30 min. The lysates were then collected and centrifuged at $14,000 \times g$ for 10 min at 4 °C to collect the supernatant. Protein concentration was determined using the Pierce BCA Protein Assay Kit (Thermo Scientific, Rockford, IL) before equivalent amounts of proteins were resolved by SDS-polyacrylamide gel electrophoresis (SDS-PAGE). The resulting protein bands were transferred onto a polyvinylidene fluoride (PVDF) membrane using the Trans-Blot SD Semi-Dry Transfer Cell (Bio-Rad, Hercules, CA). The PVDF membrane was then blocked with 1% BSA and incubated with the desired primary antibody. The membrane was then incubated with the appropriate HRP-conjugated secondary antibody before visualizing by enhanced chemiluminescence (ECL) using Amersham ECL Select (GE Healthcare, Buckinghamshire, UK) on the BioSpectrum 600 Imaging System (UVP, Upland, CA).

2.7. Reactive oxygen species (ROS) measurement assay

A549 cells were seeded into a black 96-well dish and incubated with 2',7'-dichlorofluorescein diacetate (DCFDA) for 60 min. The DCFDA solution was washed off and the cells were incubated with media without MNQ or with MNQ at concentrations of 10 or 20 μ M. The black 96-well dish was placed in the FLUOstar OPTIMA fluorescence microplate reader (BMG LABTECH, Ortenberg, Germany) with excitation and emission filters set at 495 nm and 519 nm, respectively. The fluorescence signals were collected every 30 min for 3 h.

2.8. Single cell gel electrophoresis (comet assay)

A549 cells were seeded into 60 mm dishes and treated with 10 μ M MNQ for 1 h. The cells were then trypsinized and washed with cold PBS. Comet assay was then performed using the OxiSelect Comet Assay Kit (Cell Biolabs, SD) according to the manufacturer's instruction. Briefly, harvested individual cells were mixed with molten low melt agarose and applied on the OxiSelect Comet Slide. The embedded cells were then lysed using the Lysis Buffer and treated with Alkaline Solution to relax and denature the DNA. Subsequently, electrophoresis of the samples was carried out under alkaline condition at 1 V/cm and 300 mA for 30 min. After electrophoresis, the samples were stained with Vista Green fluorescence dye for 15 min and viewed using the Eclipse Ti-E fluorescence inverted microscope (Nikon Instruments, Japan). Analysis was done using CASP comet assay software (CaspLab) and results were expressed as Tail Moment (Tail DNA% $\times$ Tail Length).

2.9. Pathway inhibition and ROS scavenging

For inhibition of JNK and p38 MAPK activations as well as ROS scavenging, A549 cells were pre-treated with 20 μ M specific chemical inhibitors (SP600125 or SB203580) for 1 h or 5 mM ROS scavenger NAC for 30 min before treatment with MNQ for indicated periods.

2.10. Statistical analysis

Results were expressed as mean values $\pm$ standard error (mean $\pm$ SE). The data were analyzed using Student's t-test for comparison between two groups and one way analysis of variance (ANOVA) followed by Student–Newman–Keuls Post Hoc test for comparison of more than two groups. A level of $p < 0.05$ was considered as significant.

3. Results

3.1. MNQ inhibits the growth of A549 cancer cells

To investigate the potential of MNQ in inhibiting the growth of lung cancer cells, A549 lung adenocarcinoma cells were treated with increasing concentrations of MNQ for 48 h, and cell growth was measured using SRB assay. As shown in Fig. 2A, MNQ exerted an anti-proliferative effect on A549 cells in a dose-dependent manner, with an IC_{50} value of 7.5 μ M. Interestingly, Fig. 2B showed that the viability of A549 cells was already reduced significantly after 24 h of exposure with 10 μ M MNQ, which was further reduced 48 h post-treatment and remained the same at 72 h post-treatment. These results demonstrated that the growth inhibitory effect of MNQ towards A549 lung adenocarcinoma cells is rapid, which occurs within 24 h of exposure.

3.2. MNQ induces A549 cell death via apoptosis

To investigate if MNQ inhibits the growth of A549 cells by inducing cell death via apoptosis, the Annexin V apoptosis assay was performed on A549 cells treated with 10 μ M MNQ for 3, 6, 9 and 12 h. The data obtained showed that 10 μ M MNQ induced apoptosis of A549 cells in a time-dependent manner (Fig. 2C). Exposure to MNQ caused the viable cell population to drop as early as 6 h post-treatment with a concomitant increase in early apoptotic cell population. This signifies that apoptosis was triggered at this time point. Cell death (late apoptosis) was detected at 9 h post-treatment. The amount of late apoptotic cells continued to increase as the treatment duration was prolonged up to 12 h. These results indicate that MNQ inhibits the growth of A549 cells by rapidly triggering the apoptotic mechanism.

3.3. MNQ induces A549 cell death independent of cell cycle disruption

To determine if MNQ induces A549 cell cycle arrest at any particular phase, flow cytometry cell cycle analysis was performed on A549 cells exposed to 10 μ M MNQ for 6 to 48 h. The results obtained showed that MNQ did not alter the cell cycle profile of A549 cells at any time point tested (Fig. 2D). This data indicated that MNQ induced apoptosis in A549 cells independent of cell cycle disruption, and its cytotoxic mechanism does not involve proteins or pathways specific to the cell cycle.

3.4. MNQ mediates apoptosis of A549 cells via ROS generation

To determine whether MNQ-induced A549 cell death is preceded by ROS generation, the level of ROS in MNQ-treated A549 cells was measured using the membrane-permeable fluorogenic probe 2',7'-dichlorodihydrofluorescein diacetate (DCFDA). The results obtained showed that intracellular H_2O_2 level in A549 rapidly increased within 3 h of exposure to MNQ, where 10 μ M MNQ increased intracellular H_2O_2 3.51-fold relative to non-treated cells, while the rise in H_2O_2 caused by 20 μ M MNQ treatment was 5.63-fold. This result obtained showed that exposure of A549 cells to MNQ caused an increase in intracellular H_2O_2 level in a time- and dose-dependent manner (Fig. 3A).

The important role played by MNQ-induced ROS generation in the apoptosis of A549 cells was further confirmed by pre-treating A549 cells with ROS scavenger N-acetyl-cysteine (NAC) before being exposed to MNQ. ROS measurement assay with DCFDA showed that intracellular

ROS levels in MNQ-exposed A549 cells pre-treated with 5 mM NAC for 30 min were reduced significantly, with DCF fluorescence readings comparable to that of non-treated cells (Fig. 3B). In addition, data obtained from flow cytometry apoptosis assay showed that NAC pre-treatment significantly reduced the early and late apoptotic cell populations of MNQ-treated A549 cells, thus rescuing the cells from MNQ-mediated cytotoxicity (Fig. 6C). These results proved the importance of ROS generation in MNQ-mediated apoptosis of A549 cells, and reduction of intracellular ROS via ROS scavenging interferes with MNQ-mediated cytotoxicity.

3.5. MNQ-induced ROS generation mediates A549 apoptosis by causing DNA damage

As oxidative DNA damage is a well-known mediator of cell death mechanism, the effect of MNQ-mediated ROS generation on the DNA of A549 was investigated. The comet assay data demonstrated that significant DNA damage was detected in A549 after exposure to MNQ for 1 h (Fig. 4). As the first sign of early apoptosis was apparent only at 6 h post-treatment, this indicated that this DNA damage was induced by the increase of intracellular ROS caused by MNQ exposure, and is not the result of DNA fragmentation catalyzed by endonucleases during apoptosis. In addition, DNA damage in A549 caused by MNQ exposure was efficiently prevented in A549 pre-treated with NAC, proving that the DNA damage was caused by oxidative stress induced by MNQ exposure.

3.6. Activations of JNK and p38 MAPK signaling pathways are required for MNQ-mediated A549 cytotoxicity

To investigate whether MNQ-mediated apoptosis in A549 cells involves the activation of the MAPK signaling pathways, A549 cells were treated with 10 μ M MNQ for various time points. Western blot results showed that MNQ triggered the activation of both JNK and p38 in A549 cells as early as 1 h post-treatment, while its effect on the activation of ERK was minimal. The increase in p-JNK and p-p38 protein levels in MNQ-treated A549 cells peaked at 3 h post-treatment, after which the levels started to decrease gradually. Nevertheless, the levels of both p-JNK and p-p38 in MNQ-treated A549 cells were still higher relative to control cells at 9 h post-treatment. Thus, the results showed that exposure of A549 cells to MNQ resulted in prolonged activation of both the JNK and p38 signaling pathways (Fig. 5).

The importance of both JNK and p38-MAPK signaling pathways was further investigated by inhibiting JNK and p38 activation with anthrapyrazolone SP600125 and pyridinyl imidazole SB203580, respectively. Western blot results showed that pre-treatment with the inhibitors for 1 h reduced the level of activated JNK or p38 in MNQ-treated cells (Fig. 6A and B), and reduction in the activation of either signaling pathway successfully reduced MNQ-mediated apoptosis in A549 (Fig. 6C). These data proved that activations of both JNK and p38 MAPK signaling pathways are required for MNQ-induced A549 apoptosis.

3.7. JNK and p38 activations are mediated by MNQ-induced oxidative stress

As both JNK and p38 signaling cascades are commonly activated by a variety of stress stimuli, we investigated whether the activation of these two MAPK pathways are due to MNQ-induced oxidative stress. Western blot data revealed that pre-treatment of A549 cells with NAC before exposure to MNQ reduced the intracellular levels of p-JNK and p-p38 in a fashion similar to that of SP600125 and SB203580 pre-treatment (Fig. 6A and B). These results showed that activations of JNK and p38 signaling pathways in MNQ-treated A549 cells are dependent on oxidative stress, and reduction of this oxidative stress negates the activation of these stress-related signaling pathways which triggers apoptosis.

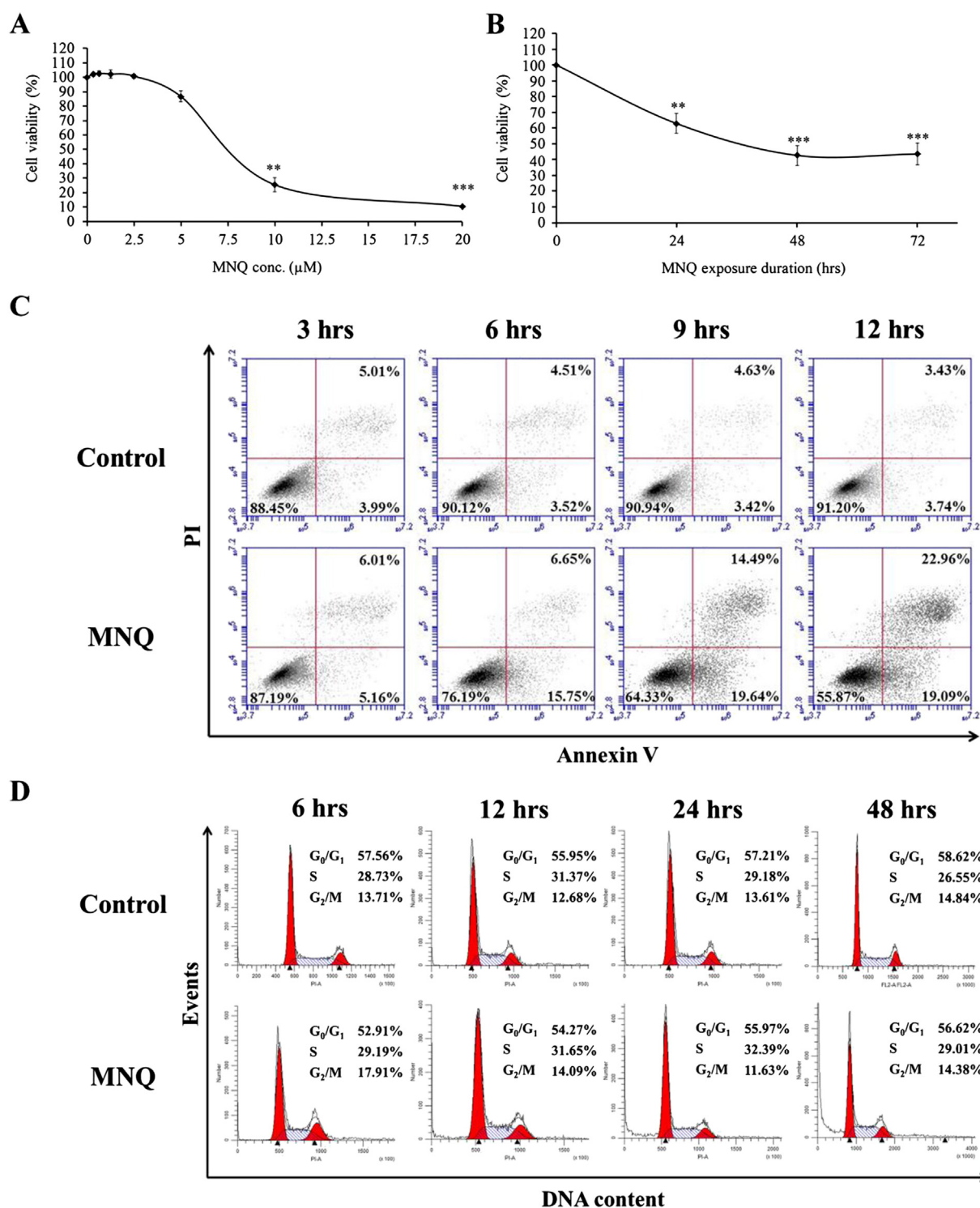


Fig. 2. MNQ inhibits the growth of A549 cells via apoptosis independent of cell cycle arrest. (A) Dose-dependent effect of MNQ on the viability of A549 cells. A549 cells were treated with increasing concentrations of MNQ for 48 h, and cell growth was measured using SRB assay. **p < 0.01 MNQ versus non-treated cells, ***p < 0.001 MNQ versus non-treated cells. (B) Time-dependent effect of MNQ on the viability of A549 cells. A549 cells were treated with 10 μM MNQ for 24, 48, and 72 h, and cell growth was measured using SRB assay. **p < 0.01 MNQ versus non-treated cells, ***p < 0.001 MNQ versus non-treated cells. (C) Apoptosis measurement of MNQ-treated A549 cells. Control and MNQ-treated A549 cells were stained with Annexin V-FITC/PI and apoptotic events were detected via flow cytometry. (D) Cell cycle analysis of MNQ-treated A549 cells. Control and MNQ-treated A549 cells were fixed, stained with PI and analyzed using flow cytometry for cell cycle distribution.

4. Discussion

Previous studies have shown that MNQ was able to inhibit the growth of cancer cells of different origins but its mechanism of action is poorly understood [14,22,41]. In this study, we demonstrated that

MNQ effectively kills A549 cells in a ROS-dependent mechanism. Our SRB assay indicated that MNQ inhibited the proliferation of A549 lung adenocarcinoma cells in a dose-dependent manner. Besides that, the growth inhibitory effect of MNQ towards A549 cells was demonstrated to be rapid, as the viability of A549 cells was significantly reduced after

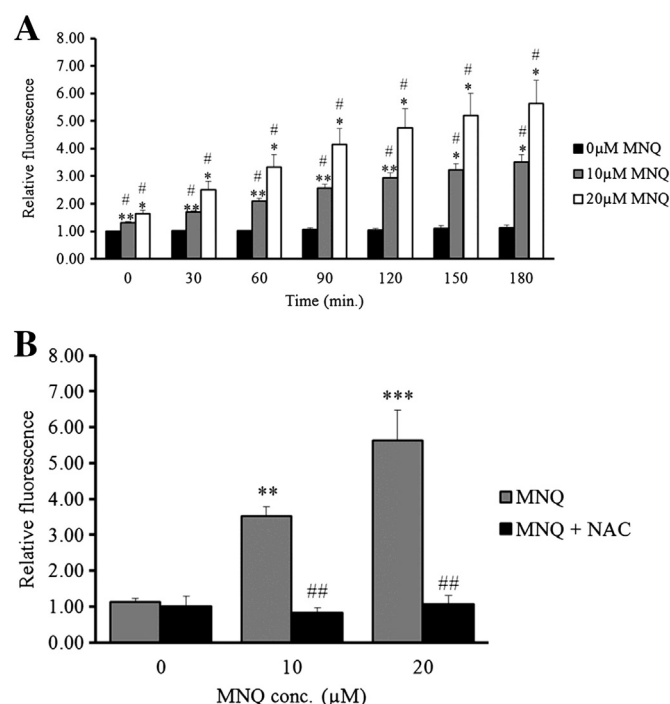


Fig. 3. MNQ induces ROS generation in A549 cells. (A) A549 cells were stained with DCFDA before exposure to MNQ. Fluorescence signals were measured every 30 min for 3 h. * $p < 0.05$, ** $p < 0.01$ versus fluorescence of control cells at 0 min. # $p < 0.05$ between 0 μ M MNQ and 10 μ M MNQ or 10 μ M MNQ and 20 μ M MNQ at each time point. (B) Pre-treating A549 cells with ROS scavenger NAC reduced the amount of ROS generated by MNQ exposure. ** $p < 0.01$, *** $p < 0.001$ MNQ vs. control. ## $p < 0.01$ MNQ vs. MNQ + NAC.

24 h of MNQ exposure. This is further supported by flow cytometry data which demonstrated that MNQ induced apoptosis in A549 cells which can be detected as early as 6 h post-treatment. Surprisingly, MNQ mediated apoptosis of A549 cells independent of cell cycle arrest. This effect differs from those exerted by several anticancer drugs such as flavopiridol, staurosporine, etoposide and paclitaxel, which target cell cycle-specific proteins to cause cancer cell death dependent of cell cycle arrest [2,4,34,36]. However, the usage of cell cycle-targeting drugs is limited as they tend to affect non-cancer cells as well, which

could be due to non-specificity of these drugs in mediating cell cycle arrest [12]. Thus our data indicated that MNQ employs a cytotoxic mechanism which is not dependent on the cell cycle.

Several studies have shown the importance of ROS in cancer cell death induced by multiple stimuli. Compounds with quinone moieties are known to undergo redox cycling to form semiquinone radicals, which may react with oxygen to produce ROS [25,37]. As MNQ is a quinone-containing compound, it has the potential to generate intracellular ROS via redox cycling. In our study, MNQ-induced apoptosis of A549 cells was shown to be dependent on ROS generation, since ROS scavenging using NAC prevented both MNQ-mediated oxidative DNA damage and apoptosis of A549. As most cancer cells possess inherently higher ROS level compared to normal cells, further induction of ROS generation using MNQ will overcome the antioxidant threshold limit of the cancer cells, thus damaging cellular components to cause cell death. In contrast, as intracellular ROS level in non-cancer cells is lower, the antioxidant defense mechanism is sufficient to deal with additional oxidative stress caused by the same xenobiotic insult, thus conferring resistance against the cytotoxic effect of the anticancer compound [24]. This difference in the redox state between cancer cells and non-cancer cells allows ROS-inducing compounds such as MNQ to better target cancer cells for killing [20].

Although DNA damage is usually associated with cell cycle arrest, this was not the case in MNQ-treated A549 cells. This could be due to rapid ROS generation which causes severe DNA damage, rendering DNA repair impossible and thus leads to the early trigger of the apoptotic cell death mechanism without cell cycle arrest. The presence of oxidative DNA damage which preludes A549 cell death after exposure to MNQ also cued the involvement of JNK and p38 MAPK signaling cascades, as these two are well known stress-induced signaling pathways [38]. Consistently, our Western blot result showed that JNK and p38 in A549 cells were activated as early as 1 h after MNQ exposure. These activations were sustained up to 6 h, coinciding with the trigger of apoptosis at that time as detected by the Annexin V apoptosis assay. ROS scavenging by NAC blocked the activation of JNK and p38 by MNQ, indicating that these two MAPKs were indeed stimulated by MNQ-mediated oxidative stress. This could be due to the lack of oxidative DNA damage in MNQ-treated A549 cells pre-treated with NAC, as DNA damage was reported to induce the activation of Threonine and One amino acid (TAO) kinases, MEK1 (MAP3K1), and transforming growth factor- β -activated kinase 1 (TAK1), which are the MAP3Ks of

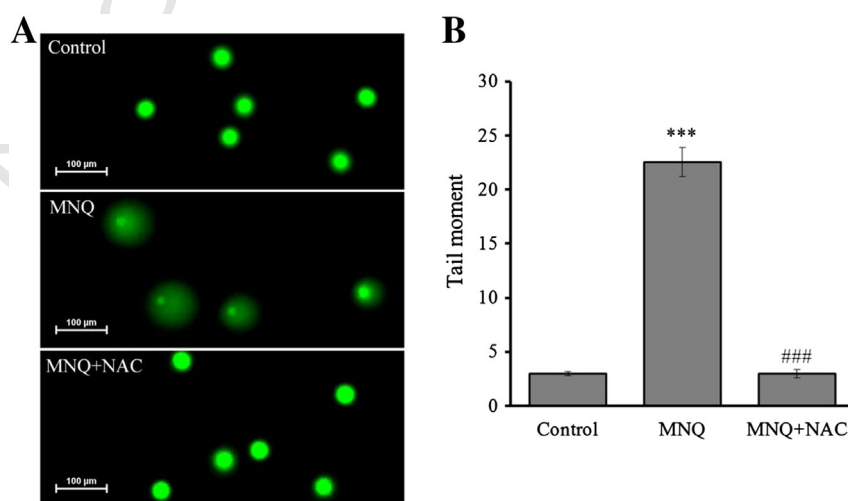


Fig. 4. MNQ-induced ROS generation leads to oxidative DNA damage. DNA damage detected by alkaline comet assay. A549 cells were treated with 10 μ M MNQ for 1 h in the presence or absence of NAC, then harvested for alkaline comet assay. (A) Representative images of the comet assay. Scale bar = 100 μ m, 200 $\times$ magnification. (B) Quantitative results of the comet assay expressed as comet tail moment (Tail DNA% $\times$ Tail Length). *** $p < 0.001$ MNQ vs. control. ### $p < 0.01$ MNQ vs. MNQ + NAC.

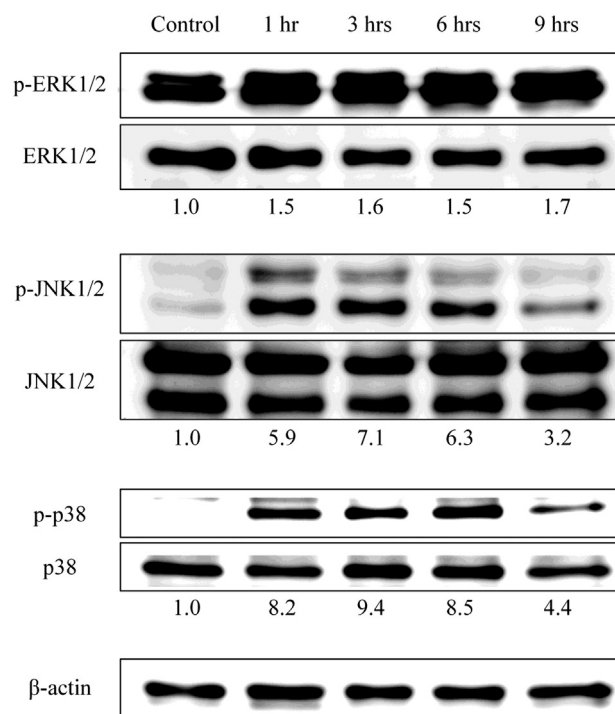


Fig. 5. MNQ induces activations of JNK and p38 MAPK signaling pathways. Whole cell extracts from control and MNQ-treated A549 cells were subjected to Western blotting using anti-p-ERK1/2, anti-p-JNK1/2, anti-p-p38, anti-ERK1/2, anti-JNK1/2, and anti-p38 antibodies. β-Actin was used as an internal control. The numbers shown represent the mean fold change of the respective phospho-protein levels in MNQ-treated cells relative to the control cells from three independent experiments.

Downregulation of JNK and p38 activation by specific chemical inhibitors decreased MNQ-mediated A549 apoptosis, indicating that MNQ requires the activations of both JNK and p38 signaling pathways in A549 cells in order to trigger the apoptotic pathway, and inhibition of either signaling pathway will prevent MNQ from killing A549 cells. This result suggests that certain distinct downstream targets of both JNK and p38 are required for the activation of the apoptotic pathway in MNQ-treated A549 cells. Similarly, studies on other anticancer compounds such as cryptotanshinone and panaxydol also reported that inhibition of either JNK or p38 pathway prevented the drugs from inducing cancer cell apoptosis [9,18]. These reports highlighted the importance of targeting JNK and p38 MAPK signaling pathways in anticancer strategy and their modulation by MNQ in A549 cells indicates the potential of the compound to be used in lung cancer management. In contrast, the ERK pathway, which is usually linked to cell survival and proliferation, was unaffected in A549 cells exposed to MNQ [27]. Thus, our results showed that MNQ-induced oxidative stress modulated the MAPK signaling pathways by steering them towards triggering cell death mechanism instead of promoting cell survival.

In conclusion, the current study demonstrated that MNQ induced apoptotic cell death of A549 lung adenocarcinoma cells by stimulating ROS generation, which caused oxidative DNA damage and subsequent activation of JNK and p38 MAPK signaling pathways. This study is the first to provide an insight into the molecular mechanism of MNQ-induced death of a lung cancer cell, suggesting that MNQ may be a promising chemotherapeutic agent against NSCLC.

Conflict of interest statement

The authors declare that there are no conflicts of interest.

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the JNK and p38 signaling pathways [17,32,43]. Thus, abrogating MNQ-mediated oxidative DNA damage via ROS scavenging would inhibit the activation of JNK and p38 signaling pathways.

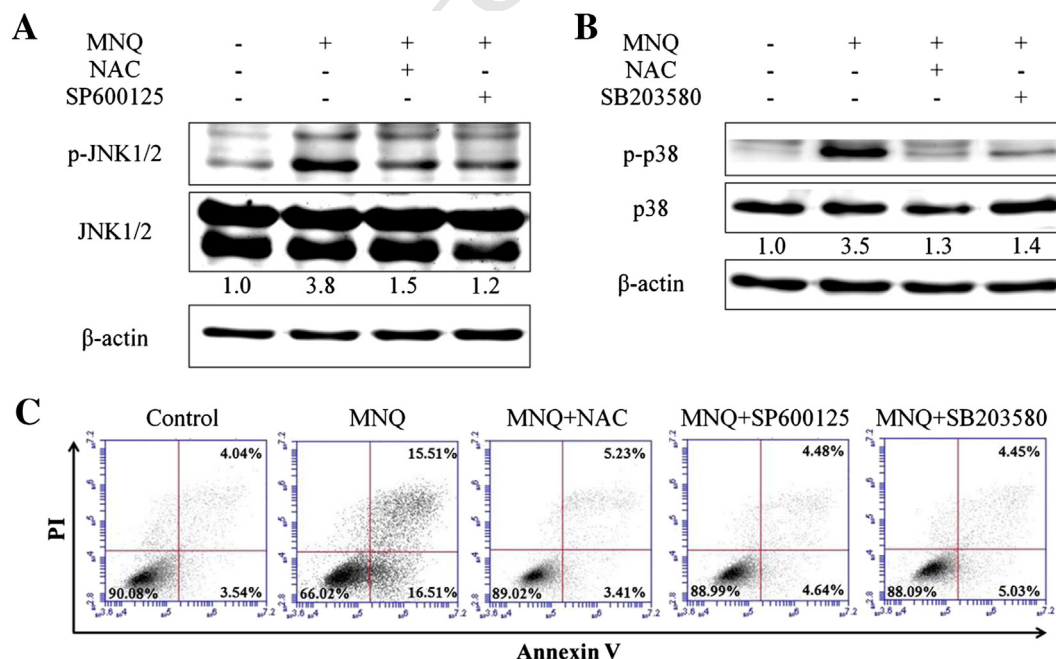


Fig. 6. MNQ-induced A549 apoptosis is dependent on ROS generation, JNK and p38 signaling pathways. (A) Inhibitory effect of NAC and SP600125 on JNK activation. A549 cells were pre-treated with NAC or SP600125 before exposure to MNQ for 3 h. Protein levels were probed via Western blotting. The numbers shown represent the mean fold change of the respective phospho-protein levels in MNQ-treated cells relative to the control cells from three independent experiments. (B) Inhibitory effect of NAC and SB203580 on p38 activation. A549 cells were pre-treated with NAC or SB203580 before exposure to MNQ for 3 h. Protein levels were probed via Western blotting. β-Actin was used as an internal control. The numbers shown represent the mean fold change of the respective phospho-protein levels in MNQ-treated cells relative to the control cells from three independent experiments. (C) Effect of NAC, SP600125 and SB203580 on MNQ-induced A549 apoptosis. A549 cells were pre-treated with NAC, SP600125 or SB203580 before being exposed to MNQ for 9 h. Apoptotic events were measured via Annexin V/PI apoptosis assay.

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